

Assignment No. 3

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

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Batch - S5

Q. Implement the database schema using SQL Data Definition Language (DDL) commands. To implement a given conceptual or logical database design by creating a relational database schema using SQL Data Definition Language (DDL) commands. You are required to translate the designed ER diagram (as part of Assignment No 2) into SQL tables while ensuring data integrity, consistency, and correctness through appropriate constraints.

The implementation must strictly follow standard relational database principles and SQL syntax.

You must perform the following tasks:

1. Table Creation
2. Key Constraints
3. Domain and Attribute Constraints
4. Relationship Handling
5. Special Design Considerations (if any)
6. Schema Validation

1. PERSON TABLE

```
CREATE TABLE Person (  
    person_id NUMBER PRIMARY KEY,  
    first_name VARCHAR2(50),  
    last_name VARCHAR2(50),  
    street_no VARCHAR2(10),  
    street_name VARCHAR2(50),  
    city VARCHAR2(50),  
    zip_code VARCHAR2(10),  
    date_of_birth DATE,  
    email VARCHAR2(100) UNIQUE  
);
```

Output-

Object Type	TABLE	Object	PERSON						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PERSON	PERSON ID	NUMBER	22	-	-	1	-	-	-
	FIRST NAME	VARCHAR2	50	-	-	-	✓	-	-
	LAST NAME	VARCHAR2	50	-	-	-	✓	-	-
	STREET NO	VARCHAR2	10	-	-	-	✓	-	-
	STREET NAME	VARCHAR2	50	-	-	-	✓	-	-
	CITY	VARCHAR2	50	-	-	-	✓	-	-
	ZIP CODE	VARCHAR2	10	-	-	-	✓	-	-
	DATE OF BIRTH	DATE	7	-	-	-	✓	-	-
	EMAIL	VARCHAR2	100	-	-	-	✓	-	-
									1 - 9

2. STUDENT TABLE (ISA Person)

CREATE TABLE Student (

student_id NUMBER PRIMARY KEY,

person_id NUMBER UNIQUE,

enrollment_date DATE,

graduation_date DATE,

semester NUMBER CHECK (semester > 0),

cgpa NUMBER(3,2) CHECK (cgpa BETWEEN 0 AND 10),

CONSTRAINT fk_student_person

FOREIGN KEY (person_id)

REFERENCES Person(person_id)

ON DELETE CASCADE

);

Output-

Object Type	TABLE	Object	STUDENT							
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment	
<u>STUDENT</u>	<u>STUDENT_ID</u>	NUMBER	22	-	-	1	-	-	-	
	<u>PERSON_ID</u>	NUMBER	22	-	-	-	✓	-	-	
	<u>ENROLLMENT_DATE</u>	DATE	7	-	-	-	✓	-	-	
	<u>GRADUATION_DATE</u>	DATE	7	-	-	-	✓	-	-	
	<u>SEMESTER</u>	NUMBER	22	-	-	-	✓	-	-	
	<u>CGPA</u>	NUMBER	-	3	2	-	✓	-	-	
									1 - 6	

3. FACULTY TABLE

```
CREATE TABLE Faculty (  
    faculty_id NUMBER PRIMARY KEY,  
    person_id NUMBER UNIQUE,  
    designation VARCHAR2(50),  
    office_location VARCHAR2(100),  
  
    CONSTRAINT fk_faculty_person  
    FOREIGN KEY (person_id)  
    REFERENCES Person(person_id)  
    ON DELETE CASCADE  
);
```

Output-

Object Type	TABLE	Object	FACULTY						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FACULTY	FACULTY_ID	NUMBER	22	-	-	1	-	-	-
	PERSON_ID	NUMBER	22	-	-	-	✓	-	-
	DESIGNATION	VARCHAR2	50	-	-	-	✓	-	-
	OFFICE_LOCATION	VARCHAR2	100	-	-	-	✓	-	-
									1 - 4

4. STAFF TABLE

```
CREATE TABLE Staff (  
    staff_id NUMBER PRIMARY KEY,  
    person_id NUMBER UNIQUE,  
    role VARCHAR2(50),  
  
    CONSTRAINT fk_staff_person  
    FOREIGN KEY (person_id)  
    REFERENCES Person(person_id)  
    ON DELETE CASCADE  
);
```

Output-

Object Type	TABLE	Object	STAFF						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STAFF	STAFF_ID	NUMBER	22	-	-	1	-	-	-
	PERSON_ID	NUMBER	22	-	-	-	✓	-	-
	ROLE	VARCHAR2	50	-	-	-	✓	-	-
									1 - 3

5. DEPARTMENT TABLE

```
CREATE TABLE Department (  
    dept_id NUMBER PRIMARY KEY,  
    dept_name VARCHAR2(100) NOT NULL,  
    est_year NUMBER CHECK (est_year > 1900),  
    building VARCHAR2(100)  
);
```

Output-

Object Type	TABLE	Object	DEPARTMENT						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>DEPARTMENT</u>	<u>DEPT_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>DEPT_NAME</u>	VARCHAR2	100	-	-	-	-	-	-
	<u>EST_YEAR</u>	NUMBER	22	-	-	-	✓	-	-
	<u>BUILDING</u>	VARCHAR2	100	-	-	-	✓	-	-
									1 - 4

6. COURSE TABLE

```
CREATE TABLE Course (  
    course_id NUMBER PRIMARY KEY,  
    title VARCHAR2(100),  
    course_code VARCHAR2(20) UNIQUE,  
    credits NUMBER CHECK (credits > 0)  
);
```

Output-

[illegible]

7. CLASSROOM TABLE

```
CREATE TABLE Classroom (  
    room_id NUMBER PRIMARY KEY,  
    capacity NUMBER CHECK (capacity > 0)  
);
```

Output-

Object Type	TABLE	Object	CLASSROOM						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>CLASSROOM</u>	<u>ROOM_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>CAPACITY</u>	NUMBER	22	-	-	-	✓	-	-
									1 - 2

8. LABORATORY TABLE

```
CREATE TABLE Laboratory (
    lab_id NUMBER PRIMARY KEY,
    lab_name VARCHAR2(100)
);
```

Output-

Object Type	TABLE	Object	LABORATORY						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>LABORATORY</u>	<u>LAB_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>LAB_NAME</u>	VARCHAR2	100	-	-	-	✓	-	-
									1 - 2

9. RESEARCH PROJECT TABLE

```
CREATE TABLE ResearchProject (
    project_id NUMBER PRIMARY KEY,
    title VARCHAR2(100),
    start_date DATE,
    end_date DATE,
    dept_id NUMBER,

    CONSTRAINT fk_project_dept
    FOREIGN KEY (dept_id)
    REFERENCES Department(dept_id)
    ON DELETE CASCADE
);
```

Output-

Object Type	TABLE Object	RESEARCHPROJECT							
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
RESEARCHPROJECT	PROJECT_ID	NUMBER	22	-	-	1	-	-	-
	TITLE	VARCHAR2	100	-	-	-	✓	-	-
	START_DATE	DATE	7	-	-	-	✓	-	-
	END_DATE	DATE	7	-	-	-	✓	-	-
	DEPT_ID	NUMBER	22	-	-	-	✓	-	-
									1 - 5

10. PUBLICATION TABLE

```
CREATE TABLE Publication (  
    pub_id NUMBER PRIMARY KEY,  
    title VARCHAR2(100),  
    DOI VARCHAR2(100) UNIQUE,  
    project_id NUMBER,  
  
    CONSTRAINT fk_pub_project  
    FOREIGN KEY (project_id)  
    REFERENCES ResearchProject(project_id)  
    ON DELETE CASCADE  
);
```

Output-

Object Type	TABLE	Object	PUBLICATION						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>PUBLICATION</u>	<u>PUB_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>TITLE</u>	VARCHAR2	100	-	-	-	✓	-	-
	<u>DOI</u>	VARCHAR2	100	-	-	-	✓	-	-
	<u>PROJECT_ID</u>	NUMBER	22	-	-	-	✓	-	-
									1 - 4

RELATIONSHIP TABLES

1) ENROLLS (M:N)

```
CREATE TABLE Enrolls (
```

student_id NUMBER,

course_id NUMBER,

PRIMARY KEY (student_id, course_id),

FOREIGN KEY (student_id)

REFERENCES Student(student_id)

ON DELETE CASCADE,

FOREIGN KEY (course_id)

REFERENCES Course(course_id)

ON DELETE CASCADE

$$);$$

Output-

Object Type	TABLE	Object	ENROLLS						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>ENROLLS</u>	<u>STUDENT_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>COURSE_ID</u>	NUMBER	22	-	-	2	-	-	-
									1 - 2

2) TEACHES

```
CREATE TABLE Teaches (
    faculty_id NUMBER,
    course_id NUMBER,
    PRIMARY KEY (faculty_id, course_id),
    FOREIGN KEY (faculty_id)
    REFERENCES Faculty(faculty_id),
    FOREIGN KEY (course_id)
    REFERENCES Course(course_id)
);
```

Output-

Object Type	TABLE	Object	TEACHES						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>TEACHES</u>	<u>FACULTY_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>COURSE_ID</u>	NUMBER	22	-	-	2	-	-	-
									1 - 2

3) WORKS_ON (Faculty → Department)

```
CREATE TABLE Works_On (
    faculty_id NUMBER,
    dept_id NUMBER,

    PRIMARY KEY (faculty_id, dept_id),

    FOREIGN KEY (faculty_id)
    REFERENCES Faculty(faculty_id),

    FOREIGN KEY (dept_id)
    REFERENCES Department(dept_id)
);
```

Output-

Object Type	TABLE	Object	WORKS_ON						
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>WORKS_ON</u>	<u>FACULTY_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>DEPT_ID</u>	NUMBER	22	-	-	2	-	-	-
									1-2

4) SCHEDULED_IN

```
CREATE TABLE Scheduled_In (
    course_id NUMBER,
    room_id NUMBER,

    PRIMARY KEY (course_id, room_id),

    FOREIGN KEY (course_id)
    REFERENCES Course(course_id),

    FOREIGN KEY (room_id)
    REFERENCES Classroom(room_id)
);
```

Output-

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>SCHEDULED_IN</u>	<u>COURSE_ID</u>	NUMBER	22	-	-	1	-	-	-
	<u>ROOM_ID</u>	NUMBER	22	-	-	2	-	-	-
									1-2

CLASSROOM HAS LAB

```
CREATE TABLE Has_Lab (  
    room_id NUMBER,  
    lab_id NUMBER,  
  
    PRIMARY KEY (room_id, lab_id),  
  
    FOREIGN KEY (room_id)  
    REFERENCES Classroom(room_id),  
  
    FOREIGN KEY (lab_id)  
    REFERENCES Laboratory(lab_id)  
);
```

Output-

Object Type	TABLE	Object	HAS_LAB							
	Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
	<u>HAS_LAB</u>	<u>ROOM_ID</u>	NUMBER	22	-	-	1	-	-	-
		<u>LAB_ID</u>	NUMBER	22	-	-	2	-	-	-
										1 - 2